

Decision-Making under Certainty

Behavioural Economics

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How to think about the BE.

- How does it relate to “standard economics”?
- Hedonic Psychology.
- Cognitive sciences and BE.
- Psychology and economics.
- Is behavioural economics really different, or is it a case of old wine in a new bottle?
 - “Behavioral economics can be said to have a short history but a long past.”

Opportunity Costs

- Investing in real estate is better than not investing.
- Is it profitable if there was an even better option available?
- Let us look at the following decision tree.

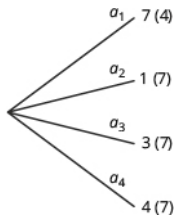


- *Explicit cost*. The upfront cost.
- The opportunity cost – or *implicit cost* – of an alternative is the value of what you would have to forgo if you choose it.

Opportunity Costs

Formally

$$c(a_i) = \max\{u(a_1), u(a_2), \dots, u(a_{i-1}), u(a_{i+1}), \dots, u(a_n)\}$$



Sunk Costs

- These are extremely important, and often lead to a behavioural mistake people make.
- This is especially something to be cautious of if you plan to have a start-up.
- Very common with R&D.
- Failing to ignore sunk costs is referred to as honouring sunk costs, or as committing the *sunk-cost fallacy*.

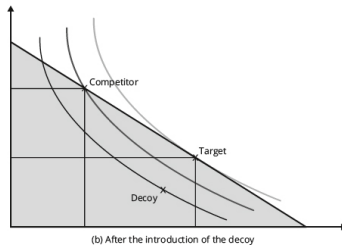
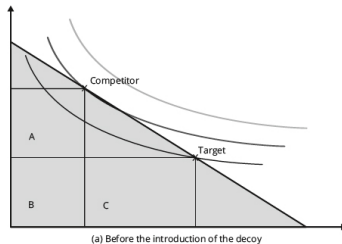
Menu dependence and the decoy effect

Expansion condition. If x is chosen from the menu $\{x, y\}$, assuming that you are not indifferent between x and y , you must not choose y from the menu $\{x, y, z\}$.

Table 3.1 *The Economist* subscription offers

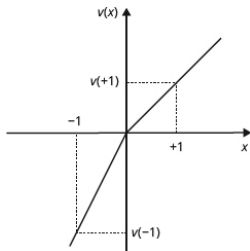
	Economist.com offers	Price
Option 1	Web subscription	\$59
Option 2	Print subscription	\$125
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Decoy Effect



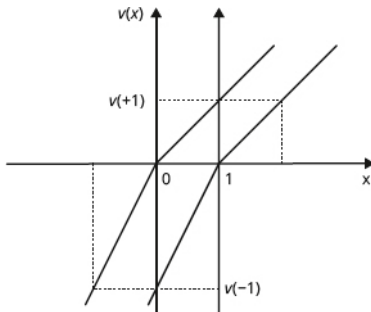
Loss Aversion and the Endowment Effect

- Willingness to pay (WTP) vs Willingness to accept (WTA).
- Endowment effect or reference-point phenomena framing effects.
- *Loss aversion*. The apparent fact that people dislike losses more than they like commensurate gains.

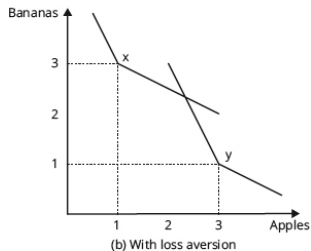
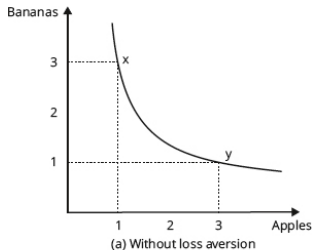


Value Function

- Behavioral economists capture loss aversion by means of a *value function* $v(\cdot)$, which represents how an agent evaluates a change.



Indifference Curves with Loss Aversion



Anchoring and Adjustment

Example 3.52 Africa and the UN. Spin a wheel of fortune to come up with a number between 0 and 100, and invite your participants to answer the following two questions:

- (a) Is the percentage of African nations in the United Nations (UN) greater than or less than the number?
 - (b) What is the actual percentage of African nations in the UN?
- We make judgements not by actually computing probabilities and utilities but by following *heuristics*.
 - The behavior pattern can be said to violate *procedure invariance*.